Executive tear sheet · Serendib University Endowment Fund

Final model v6: monthly dynamic risk parity across SPY, AGG, GLD and XLE with a recession overlay · USD 9.6m invested of the USD 10m fund after the first-year cash reserve · data 3 Jan 2005 to 30 Sep 2026 · prepared 31 Oct 2026 · Team Ghosted (solo entry) · Quant Edge 1.0 final

Recommendation: hold this portfolio (v6, the owner's final model). From 1 October 2026 hold SPY 20.8%, AGG 51.7%, GLD 8.4%, XLE 19.1%. At every month-end close reset to equal-risk weights (EWMA covariance, λ 0.97, 750 days), and while the yield curve (10-year minus 3-month) has been inverted in the last 24 months *and* SPY closes below its 12-month average, hold half of SPY and XLE in cash; the overlay is **off** at the 30 Sep 2026 close. Monitor one-day 99% VaR daily with GARCH-EVT (latest USD 110,153; FHS USD 114,279), using the higher of the two in any month the overlay switches. Call an extra committee review if the fall from the previous high passes 15% or 99% VaR breaches exceed 9 in any 250 days.

Disclosed: under the team's pre-registered rule v6 is *evaluation incomplete*: its VaR breach counts match the 99% level but breaches cluster (conditional coverage p 0.011 GARCH-EVT, 0.009 FHS). The owner adopted it for its smaller falls; the v5 core passes with GARCH-EVT (p 0.091). **Main risk accepted:** a crash before the next month-end meets today's weights; held through the GFC and COVID they would have fallen 24.5% and 20.9%. Team proposal; independent human review pending.

Allocation to hold now

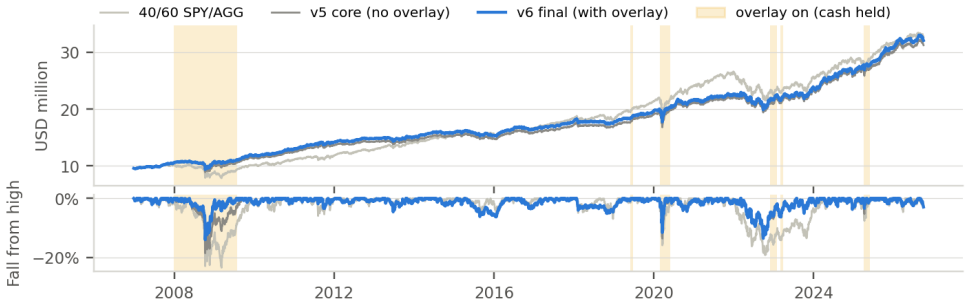
ETF	Exposure	Weight	USD	Vol. share	ES99 share
SPY	US large-cap equity	20.8%	1,997,059	25.0%	37.4%
AGG	US investment-grade bonds	51.7%	4,965,293	25.0%	6.2%
GLD	Gold	8.4%	802,656	25.0%	3.6%
XLE	US energy equity	19.1%	1,834,992	25.0%	52.8%
Total	cash 0.0%	100.0%	9,600,000	100.0%	100.0%

XLE is 19.1% of the money but 52.8% of 99% ES: equal volatility shares are not equal tail shares. The overlay held cash in 28 of 237 months: Dec 2007–Jun 2009; May 2019; Feb 2020–Apr 2020; Nov 2022–Dec 2022; Feb 2023; Mar 2025–Apr 2025.

Performance, 28 Dec 2006 to 30 Sep 2026

KPI	v6 final	v5 core	40/60
CAGR (real)	6.30% (3.61%)	6.16% (3.48%)	6.40% (3.71%)
Volatility	6.0%	6.6%	8.4%
Maximum drawdown	13.9%	18.5%	23.3%
Sharpe; Sortino	0.79; 1.49	0.71; 1.35	0.60; 1.12
Calmar; beta to SPY	0.45; 0.16	0.33; 0.21	0.27; 0.39
Jensen's alpha	3.01%	2.43%	0.78%
Longest under water	349 days	346 days	564 days

Daily simple returns, zero trading cost, before the cash reserve; overlay cash earns 0%. Real deflates by the policy CPI rate (2.59% a year, US CPI 2005-2026). Sharpe over the 3-month T-bill.



Growth of USD 9.6m and fall from the previous high: v6, v5 core and 40/60 SPY/AGG, daily, zero cost, before the cash reserve. Orange: months the overlay held cash.

Risk: one day, NAV USD 9.6m, today's weights

Model, 99%	VaR (USD)	ES (USD)
Historical simulation	155,142	249,995
Parametric (normal)	131,589	150,975
Monte Carlo (Student-t)	150,365	215,225
EVT (POT-GPD)	159,994	245,575
GARCH-EVT (current volatility)	121,525	153,436

ES 95% bootstrap interval USD 182,694 to USD 332,093. Historical ES is 1.66× the normal model's: fat tails. Fitted on 2005-2026 with today's weights.

Validation: 99% VaR backtest, 2017 to Sep 2026

Model	Breaches / exp.	Kupiec p	Indep. p	Cond. cov. p	Last 250
v6 GARCH-EVT	29 / 24.5	0.373	0.004	0.011	green
v6 FHS	33 / 24.5	0.101	0.009	0.009	green
v5 core GARCH-EVT	26 / 24.5	0.761	0.030	0.091	green

2,449 sequential forecasts, each from data to the day before, on each path's own returns (v6 hash e3acccc6). v6 breach counts are right but cluster around overlay switches, hence the higher-of-two rule. Basel zone: green 0-4, yellow 5-9, red 10+ breaches in 250 days.

Stress tests on USD 9.6m

Scenario	Dates or shock	v6 return	v6 loss (USD)	v6 max fall	v5 core	40/60	Target held: max fall
GFC	9 Oct 2007 – 9 Mar 2009	1.8%	−171,605	13.9%	−5.0%	−23.1%	24.5%
COVID	19 Feb 2020 – 23 Mar 2020	−7.3%	703,288	11.5%	−9.5%	−15.1%	20.9%
Rates 2022	3 Jan 2022 – 12 Oct 2022	−10.0%	958,319	13.5%	−10.0%	−18.4%	12.7%
Hypothetical	SPY −25%, AGG −10%, GLD 0%, XLE −34.1%	−16.9%	1,621,961	one step			

Paths as run, month-end resets and overlay included, from the start close to the end close; a negative loss is a gain. 'Target held' buys today's weights at the start and lets them drift. The hypothetical shock is deterministic (XLE from its conditional mean in SPY's worst 10% of days, 2005-2016) and carries no probability.

The committee's three questions

What can go wrong? Stocks and bonds fall together, as in 2022 (−10.0%, the worst crisis as run; the overlay was off); a shock lands between month-ends (COVID: −7.3% in five weeks); energy reverses (XLE carries 52.8% of ES); the overlay gives false alarms or reacts late to fast crashes.

How bad can losses get? About 1 day in 100 loses more than USD 155,142; those days average USD 249,995. Worst crisis as run USD 958,319; hypothetical shock USD 1,621,961. None of these is a maximum.

How confident are we? Less than for v5: v6's risk forecasts have the right breach count but clustered breaches, and the strategy and overlay were chosen after many rules were compared on the same 2005-2026 data, so there is no untouched test period. The overlay's GFC gain (1.8% against −5.0% for v5) rests on one deep recession.

Spending, 25 years: resampled v6 history pays every reserve in 98.1% of paths but keeps USD 10m in real terms in only 19.8% (median USD 6.26m): the reserve rising 3.64% a year slowly draws down real capital. Plan spending below the 3.6% real earned historically.

Evaluation labels: performance, comparisons and stress tests are a **retrospective portfolio evaluation**; VaR backtests are a **sequential risk-forecast evaluation**; no result is an untouched out-of-sample test. Data: Yahoo Finance adjusted closes, FRED DGS10, DTB3 and CPIAUCSL. Cash is outside the brief's seven ETFs: a disclosed policy choice. AI use: Codex and Claude Code wrote the code and documents; automated checks are not human validation. Human reviewer and date: pending.